Vincent Sun

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Education

University of Waterloo Sep. 2023 – Apr. 2028
Bachelor of Computer Science
- cGPA: 91.63% (3.95/4.00), Awards: René Descartes National Scholarship, President's Scholarship of Distinction

Technical Skills

Certifications: AWS Certified Cloud Practitioner
Languages and Developer Tools: Python, JavaScript, Go, Java, BASH, C/C++, SQL, Claude Code, Codex
Frameworks/Libraries: React.js, Spring, Node.js, Django, Prisma, Boto3, React-Query, Tailwind, Next.js, OpenCV
Technologies: AWS, PostgreSQL, Snowflake, Docker, Kubernetes, GCP, Prometheus, Temporal, Argo, Elasticsearch

Work Experience

Backend Software Engineering Intern May 2026 – Present
Ramp *New York, New York*
- Building a new workflow simulation platform for approval systems that evaluates code nodes using **LLMs**, solves workflow pathing, and dynamically generates input schema, improving debugging and validation of complex logic
- Developed an automated **Temporal** cron job that uses **Sownflake** and Slack integration to detect and alert on high-complexity workflows, mitigating performance degradation and lowering UI latency

Cloud Developer Intern Sep. 2025 – Dec. 2025
SAP *Waterloo, Ontario*
- Optimized **cloud storage infrastructure** by allocating **300,000+** customer instances to shared host **GCP NetApp** and **AWS EFS** volumes, consolidating data and access points, and cutting provisioning costs by **46%**
- Developed automated **Argo** workflows to handle soft and hard deletion of **Vault** credentials, and implemented cross-region state duplication using **HANA DB** with fault-tolerant retry logic ensuring **99.9%** reliability
- Accelerated environment variable rotation by **55%** by refactoring **Helm** templates to support multi-processing
- Created **Prometheus** metrics to alert and monitor **100%** of **Argo** workflows, reducing incident detection time
- Introduced versioning to **Kubernetes** operator updates to enable synchronized changes across instances

Software Engineering Intern Jan. 2025 – Apr. 2025
MedMe Health (YC W21) *Toronto, Ontario*
- Implemented event-driven **Twilio** email system built with **AWS SQS** and **Lambda** to replace **Graphile** job processing, reducing latency by over **60%** during high-traffic periods and improving error-handling using a DLQ
- Optimized **Spring** endpoint query speeds by **79%** through data loaders and all-or-none data transactions
- Added **Prometheus** metrics and built **5** custom dashboards to monitor **DynamoDB** usage with alert thresholds
- Created cookie management script and refactored **webpack** build to block **100%** of third-party script injections

Software Engineering Intern May 2024 – Aug. 2024
Atlassian *Toronto, Ontario*
- Engineered a ranking algorithm using dynamic weights with **Elasticsearch** to personalize user search results, leading to a reduction in average search rank of desired results by **23%** and increased click-through rate by **8%**
- Decreased integration query time by **47%** through optimization of **Django ORM** queries to a **Postgres** database
- Created analytics widgets for users to visualize their data trends and optimized metric update times by **32%**

Projects

LeetDance (Hack the North 2025) | *Typescript, Python, React.js, Next.js, Gemini API, MediaPipe, OpenCV*
- Built a dance analysis platform that provides frame-level feedback on movement accuracy with reference videos
- Implemented pose estimation using **MediaPipe** to extract **30+** skeletal keypoints and used **OpenCV** to analyze and visualize joint deviations with timestamped feedback generation using **Gemini** for natural language coaching
- Designed a **FastAPI** backend to handle efficient data transfer and containerized using **Docker** for consistency

Chameleon (Hack the North 2024) | *Typescript, Python, React.js, OpenAI API, Tailwind, FastAPI, Axios*
- Created an AI-powered web design tool that converts user-provided images into customized frontend pages
- Developed end-to-end pipeline to transform design images into structured **JSON** schemas using **GPT-4o** and implemented recursive **Typescript** renderer to dynamically generate **20+** interactive **React** component trees
- Added iterative natural language editing system allowing users to refine UI and reduce manual redesign time